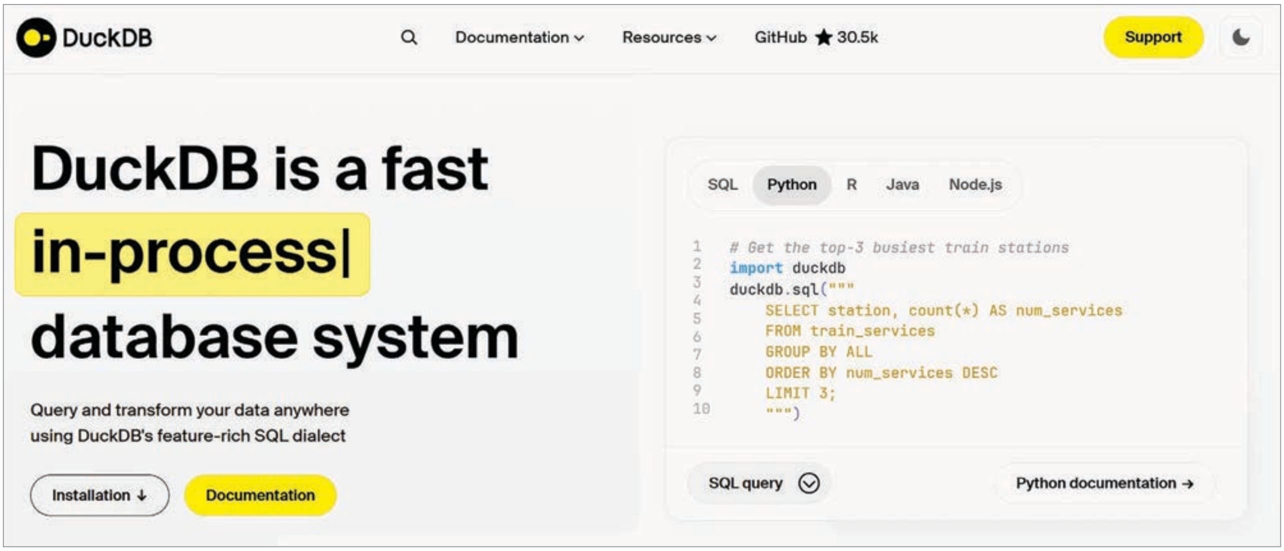

Figure 2: DuckDB in-process database for data science and analytical processes

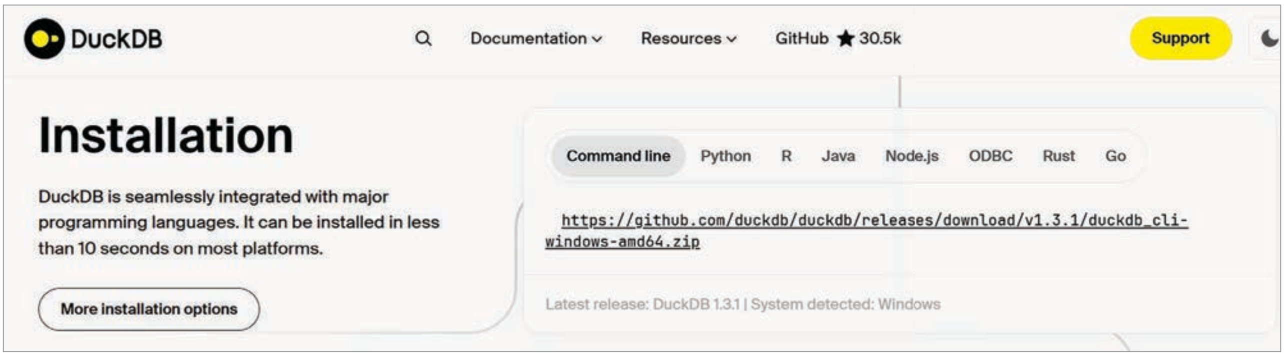

Figure 3: Installation of DuckDB for different interfaces

The software library is installed automatically and can be called in Python code.

The quick install instructions are given at *https://duckdb.org/#quickinstall*.

Table 5: Comparing DuckDB with Pandas and SQLite

Feature	DuckDB	Pandas	SQLite
SQL support	Full ANSI	None	Limited
File access	Parquet, CSV	CSV, JSON	Only CSV
OLAP focus	Yes (Columnar)	No	No (Row-based)
In-memory query	Yes	Yes	Yes
Speed (OLAP)	Very fast	Medium	Slow
Server needed	No	No	No

As can be seen in Table 5, DuckDB scores over Pandas and SQLite on all accounts. This database can be used in real-time projects associated with data science, analytics and live visualisations.

Here's an example of database operations (insert, delete, update) on sample cybersecurity data in DuckDB:

```
import duckdb
# Connect to DuckDB in memory (or use a file: duckdb.
connect("cybersecurity.db"))
duckdbconnection = duckdb.connect()

# Phase 1: Create table for suspected login attempts
duckdbconnection.execute("""
CREATE TABLE suspected_logins (
    id INTEGER PRIMARY KEY,
    user TEXT,
    remote_ip TEXT,
    login_time TIMESTAMP,
    remote_location TEXT,
    risk_factor_score FLOAT
);
""")
print("Database Table created.\n")
```